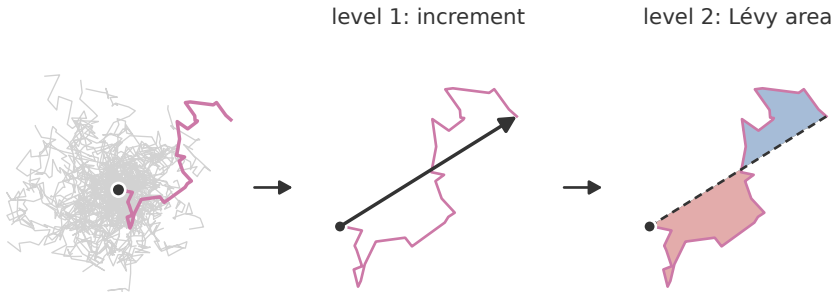

$$S_{0,t}^{i_1 \cdots i_k} = \int_{0 < s_1 < \cdots < s_k < t} dX_{s_1}^{i_1} \cdots dX_{s_k}^{i_k}$$